

Mini Project 1

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What is the question?

Are word-frequency patterns more similar between books by the same author than between books by different authors, and which words distinguish Lewis Carroll from L. Frank Baum?

Books analyzed:

- Alice's Adventures in Wonderland
- Through the Looking-Glass
- The Wonderful Wizard of Oz
- The Marvelous Land of Oz

What was the approach?

- Download the plain-text versions of the books from Project Gutenberg.
- Remove the Gutenberg headers and footers.
- Convert the text to lowercase and separated into words.
- Calculate the frequencies per 10,000 words to account for different book lengths.
- Compared content words and function words separately.
- Used bar charts and cosine similarity to compare the books.

What problems did I encounter?

- Downloaded an audiobook index instead of full text on accident.
- Jupyter kernel got stuck and needed manual restart.
- Character names and shared settings affect the content-word results.
- Similarities may come from both author style and shared subject matter due to each author's book belonging to the same series.

What results did I get?

I compared the frequencies of 20 common function words. A cosine similarity closer to 1 means that two books use these words in more similar proportions.

- Average same-author similarity: 0.994
- Average different-author similarity: 0.964
- Both same-author pairs scored higher than every different-author pair.
- This supports the hypothesis that an author's word-use patterns remain relatively consistent across books.

Comparison	Similarity
Carroll vs. Carroll	0.995
Baum vs. Baum	0.993
Different Authors	0.952-0.976

Results Cont.

Content-word results

- Carroll's books frequently used alice, queen, know, and like.
- Baum's books frequently used dorothy, tip, scarecrow, woodman, and oz.
- These words mostly reflect characters and settings, while function words provide better evidence of writing style.

These results do support my hypothesis but they do not prove that author alone caused the differences because each pair of books also shares characters, settings, and are in a common series.

What new ideas did this generate?

- I should remove the character names and locations.
- Compare unrelated books by each author and not use books from the same series.
- More authors and books to see whether the results still hold.
- Separating the books into the smaller sections and test whether the function words can identify the author of an unknown passage.